Written Assignment 1

Due: Friday 03/27/2026 at 11:59pm EST

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Disclaimer

Written assignments must be typeset in LaTeX. I have released the `.tex` files that were used to generate this pdf which you are welcome to use as the foundation of your solutions. These questions are **proof** questions, meaning that you will need to write a two-column proof with mathematical steps on the left column, and english justification on the right column. Show all of your steps.

Note that you are **NOT** allowed to use any help from LLMs, online solutions, old solutions, etc. when solving these problems. Your solutions are your own. You **are** allowed to chat with your classmates, but not with detail granular enough to copy each others work.

Question 1: Useful Convolution Properties (30 points)

Consider the convolution between two 1-dimensional functions f and g:

$$h(t) := (f * g)(t) = \int_{-\infty}^{+\infty} f(x)g(t-x)\,dx$$

Please show:

1. $f * g = g * f$
2. $f * (g * h) = (f * g) * h$
3. $f * (g + h) = (f * g) + (f * h)$
4. $\frac{d}{dt}(f * g) = (\frac{d}{dt}f) * g$ and $\frac{d}{dt}(f * g) = f * (\frac{d}{dt}g)$

Assumption. Throughout Question 1, assume the integrals converge absolutely and that the functions are regular enough for substitutions, Fubini's theorem, and differentiation under the integral sign to be valid.

1.1 $f * g = g * f$

Mathematical Steps	**Justification**
$(f * g)(t) = \int_{-\infty}^{+\infty} f(x)g(t-x)\,dx$	Definition of convolution.
$(f * g)(t) = \int_{+\infty}^{-\infty} f(t-y)g(y)(-dy)$	Substitute $y = t - x$, so $x = t - y$ and $dx = -dy$.
$(f * g)(t) = \int_{-\infty}^{+\infty} g(y)f(t-y)\,dy$	Reverse the limits and reorder the factors.

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Mathematical Steps	**Justification**
$(f * g)(t) = (g * f)(t)$	Definition of convolution again.

1.2 $f * (g * h) = (f * g) * h$

Mathematical Steps	**Justification**
$(f * (g * h))(t) = \int_{-\infty}^{+\infty} f(x)(g * h)(t - x)\, dx$	Definition of convolution.
$(f * (g * h))(t) = \int_{-\infty}^{+\infty} f(x) \left[\int_{-\infty}^{+\infty} g(y)h((t - x) - y)\, dy \right] dx$	Expand $(g * h)(t - x)$.
$(f * (g * h))(t) = \int_{-\infty}^{+\infty} f(x) \left[\int_{-\infty}^{+\infty} g(z - x)h(t - z)\, dz \right] dx$	In the inner integral, substitute $z = x + y$, so $y = z - x$ and $dy = dz$.
$(f * (g * h))(t) = \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(x)g(z - x)\, dx \right] h(t - z)\, dz$	Apply Fubini's theorem and regroup the terms.
$(f * (g * h))(t) = \int_{-\infty}^{+\infty} (f * g)(z)h(t - z)\, dz$	Definition of $(f * g)(z)$.
$(f * (g * h))(t) = ((f * g) * h)(t)$	Definition of convolution.

1.3 $f * (g + h) = (f * g) + (f * h)$

Mathematical Steps	**Justification**
$(f * (g + h))(t) = \int_{-\infty}^{+\infty} f(x)(g + h)(t - x)\, dx$	Definition of convolution.
$(f * (g + h))(t) = \int_{-\infty}^{+\infty} f(x)\big(g(t - x) + h(t - x)\big)\, dx$	Definition of pointwise addition.
$(f * (g + h))(t) = \int_{-\infty}^{+\infty} f(x)g(t - x)\, dx + \int_{-\infty}^{+\infty} f(x)h(t - x)\, dx$	Linearity of the integral.
$(f * (g + h))(t) = (f * g)(t) + (f * h)(t)$	Definition of convolution.

1.4 $\frac{d}{dt}(f * g) = (f' * g) = f * g'$

Mathematical Steps	**Justification**
$\frac{d}{dt}(f * g)(t) = \frac{d}{dt} \int_{-\infty}^{+\infty} f(x)g(t - x)\, dx$	Start from the convolution formula.
$\frac{d}{dt}(f * g)(t) = \int_{-\infty}^{+\infty} f(x) \frac{d}{dt} g(t - x)\, dx$	Differentiate under the integral sign.
$\frac{d}{dt}(f * g)(t) = \int_{-\infty}^{+\infty} f(x)g'(t - x)\, dx = (f * g')(t)$	Since $\frac{d}{dt} g(t - x) = g'(t - x)$.

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Mathematical Steps	**Justification**
$(f * g)(t) = \int_{-\infty}^{+\infty} g(y) f(t-y)\, dy$	Rewrite the convolution in the commuted form from part 1.1.
$\frac{d}{dt}(f * g)(t) = \int_{-\infty}^{+\infty} g(y) \frac{d}{dt} f(t-y)\, dy$	Differentiate under the integral sign again.
$\frac{d}{dt}(f * g)(t) = \int_{-\infty}^{+\infty} g(y) f'(t-y)\, dy = (f' * g)(t)$	Since $\frac{d}{dt} f(t-y) = f'(t-y)$.
$\therefore\ \frac{d}{dt}(f * g) = (f' * g) = f * g'$	Both required identities hold.

Question 2: Useful Laplace Transform Properties (35 points)

Before we start considering the Fourier transform of a function $f(t)$, an easier transform is called the *Laplace* transform. It is defined as follows:

$$F(s) = \mathcal{L}\{f(t)\} := \int_0^{+\infty} f(t)e^{-st}dt$$

Please show:

1. $\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$
2. $\mathcal{L}\{\frac{d^n}{dt^n}f(t)\} = s^n\mathcal{L}\{f(t)\} - \sum_{i=1}^{n} s^{n-i}\left(\frac{d^{i-1}}{dt^{i-1}}f(t)\right)\Big|_{t=0}$ where $f|_c := f(c)$ and $\frac{d^0}{dx^0}f(x) := f(x)$.
3. $\frac{d^n}{ds^n}\mathcal{L}\{f(t)\} = (-1)^n\mathcal{L}\{t^n f(t)\}$.
4. for two functions f and g show that $\mathcal{L}\{f\}\mathcal{L}\{g\} = \mathcal{L}\{f * g\}$

Assumption. Throughout Question 2, assume f, g, and the required derivatives are of exponential order so the Laplace transforms exist, the boundary terms $\lim_{t\to+\infty} e^{-st}f^{(k)}(t)$ vanish, and differentiation under the integral sign is valid. For part 2.4, use the standard causal convolution

$$(f * g)(t) := \int_0^t f(\tau)g(t-\tau)\,d\tau,$$

which is the same as the full-line convolution when f and g are zero for $t < 0$.

2.1 $\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$

Mathematical Steps	**Justification**
$\mathcal{L}\{f+g\} = \int_0^{+\infty} (f(t)+g(t))e^{-st}\,dt$	Definition of the Laplace transform.
$\mathcal{L}\{f+g\} = \int_0^{+\infty} f(t)e^{-st}\,dt + \int_0^{+\infty} g(t)e^{-st}\,dt$	Linearity of the integral.
$\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$	Definition of $\mathcal{L}$.

2.2 $\mathcal{L}\{f^{(n)}\} = s^n\mathcal{L}\{f\} - \sum_{i=1}^{n} s^{n-i}f^{(i-1)}(0)$

Mathematical Steps	**Justification**
$\mathcal{L}\{f'(t)\} = \int_0^{+\infty} f'(t)e^{-st}\,dt$	Start with the case $n = 1$.
$\mathcal{L}\{f'(t)\} = \left[f(t)e^{-st}\right]_0^{+\infty} + s\int_0^{+\infty} f(t)e^{-st}\,dt$	Integration by parts with $u = e^{-st}$, $dv = f'(t)\,dt$.
$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$	The term at $+\infty$ vanishes, and $\left[f(t)e^{-st}\right]_0^{+\infty} = 0 - f(0)$.
$\mathcal{L}\{f^{(n)}(t)\} = s^n\mathcal{L}\{f(t)\} - \sum_{i=1}^{n} s^{n-i}f^{(i-1)}(0)$	Induction hypothesis.

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Mathematical Steps	**Justification**
$\mathcal{L}\{f^{(n+1)}(t)\} = \mathcal{L}\{(f^{(n)}(t))'\} = s\mathcal{L}\{f^{(n)}(t)\} - f^{(n)}(0)$	Apply the $n = 1$ result to the function $f^{(n)}$.
$\mathcal{L}\{f^{(n+1)}(t)\} = s\left[s^n\mathcal{L}\{f(t)\} - \sum_{i=1}^{n} s^{n-i} f^{(i-1)}(0)\right] - f^{(n)}(0)$	Substitute the induction hypothesis.
$\mathcal{L}\{f^{(n+1)}(t)\} = s^{n+1}\mathcal{L}\{f(t)\} - \sum_{i=1}^{n} s^{n+1-i} f^{(i-1)}(0) - f^{(n)}(0)$	Distribute the factor s.
$\mathcal{L}\{f^{(n+1)}(t)\} = s^{n+1}\mathcal{L}\{f(t)\} - \sum_{i=1}^{n+1} s^{n+1-i} f^{(i-1)}(0)$	Absorb the last term into the sum as the $i = n + 1$ term.
$\therefore\ \mathcal{L}\{f^{(n)}(t)\} = s^n\mathcal{L}\{f(t)\} - \sum_{i=1}^{n} s^{n-i} f^{(i-1)}(0)$	The statement holds for all $n \geq 1$ by induction.

2.3 $\frac{d^n}{ds^n}\mathcal{L}\{f\} = (-1)^n\mathcal{L}\{t^n f\}$

Mathematical Steps	**Justification**
$\frac{d^n}{ds^n}\mathcal{L}\{f(t)\} = \frac{d^n}{ds^n}\int_0^{+\infty} f(t)e^{-st}\,dt$	Start from the definition of $\mathcal{L}$.
$\frac{d^n}{ds^n}\mathcal{L}\{f(t)\} = \int_0^{+\infty} f(t)\frac{\partial^n}{\partial s^n}e^{-st}\,dt$	Differentiate under the integral sign.
$\frac{d^n}{ds^n}\mathcal{L}\{f(t)\} = \int_0^{+\infty} f(t)(-t)^n e^{-st}\,dt$	$\frac{\partial^n}{\partial s^n}e^{-st} = (-t)^n e^{-st}$.
$\frac{d^n}{ds^n}\mathcal{L}\{f(t)\} = (-1)^n\int_0^{+\infty} t^n f(t)e^{-st}\,dt$	Factor out $(-1)^n$.
$\frac{d^n}{ds^n}\mathcal{L}\{f(t)\} = (-1)^n\mathcal{L}\{t^n f(t)\}$	Definition of the Laplace transform.

2.4 $\mathcal{L}\{f\}\mathcal{L}\{g\} = \mathcal{L}\{f * g\}$

Mathematical Steps	**Justification**
$\mathcal{L}\{f\}\mathcal{L}\{g\} = \left(\int_0^{+\infty} f(\tau)e^{-s\tau}\,d\tau\right)\left(\int_0^{+\infty} g(\sigma)e^{-s\sigma}\,d\sigma\right)$	Write both Laplace transforms from the definition.
$\mathcal{L}\{f\}\mathcal{L}\{g\} = \int_0^{+\infty}\int_0^{+\infty} f(\tau)g(\sigma)e^{-s(\tau+\sigma)}\,d\sigma\,d\tau$	Multiply the integrals and combine them using Fubini's theorem.
$\mathcal{L}\{f\}\mathcal{L}\{g\} = \int_0^{+\infty}\int_\tau^{+\infty} f(\tau)g(t-\tau)e^{-st}\,dt\,d\tau$	Substitute $t = \tau + \sigma$, so $\sigma = t - \tau$ and $d\sigma = dt$.
$\mathcal{L}\{f\}\mathcal{L}\{g\} = \int_0^{+\infty} e^{-st}\left[\int_0^t f(\tau)g(t-\tau)\,d\tau\right]dt$	Swap the order of integration over the region $0 \leq \tau \leq t < +\infty$.

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Mathematical Steps	**Justification**
$\mathcal{L}\{f\}\mathcal{L}\{g\} = \int_0^{+\infty} (f * g)(t) e^{-st}\, dt$	The bracketed term is exactly the causal convolution $(f * g)(t)$.
$\mathcal{L}\{f\}\mathcal{L}\{g\} = \mathcal{L}\{f * g\}$	Definition of the Laplace transform.

Question 3: Useful Fourier Transform Properties (35 points)

Now considering the Fourier transform of a function $f(t)$:

$$F(u) = \mathcal{F}\{f(t)\} := \int_{-\infty}^{+\infty} f(t)e^{-iut}dt$$

and its inverse transform:

$$f(t) = \mathcal{F}^{-1}\{F(u)\} := \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(u)e^{iut}dt$$

Please show:

1. $\mathcal{F}\{f+g\} = \mathcal{F}\{f\} + \mathcal{F}\{g\}$
2. $\mathcal{F}\{\frac{d^n}{dt^n}f(t)\} = (iu)^n\mathcal{F}\{f(t)\}$.
3. $\frac{d^n}{du^n}\mathcal{F}\{f(t)\} = (-i)^n\mathcal{F}\{t^n f(t)\}$.
4. for two functions f and g show that $\mathcal{F}\{f\}\mathcal{F}\{g\} = \mathcal{F}\{f * g\}$

Assumption. Throughout Question 3, assume the functions and required derivatives are absolutely integrable, the boundary terms $\lim_{t\to\pm\infty} f^{(k)}(t)e^{-iut}$ vanish, and Fubini's theorem plus differentiation under the integral sign may be used.

3.1 $\mathcal{F}\{f+g\} = \mathcal{F}\{f\} + \mathcal{F}\{g\}$

Mathematical Steps	**Justification**
$\mathcal{F}\{f+g\} = \int_{-\infty}^{+\infty} (f(t)+g(t))e^{-iut}\,dt$	Definition of the Fourier transform.
$\mathcal{F}\{f+g\} = \int_{-\infty}^{+\infty} f(t)e^{-iut}\,dt + \int_{-\infty}^{+\infty} g(t)e^{-iut}\,dt$	Linearity of the integral.
$\mathcal{F}\{f+g\} = \mathcal{F}\{f\} + \mathcal{F}\{g\}$	Definition of $\mathcal{F}$.

3.2 $\mathcal{F}\{f^{(n)}\} = (iu)^n\mathcal{F}\{f\}$

Mathematical Steps	**Justification**
$\mathcal{F}\{f'(t)\} = \int_{-\infty}^{+\infty} f'(t)e^{-iut}\,dt$	Start with the case $n = 1$.
$\mathcal{F}\{f'(t)\} = \left[f(t)e^{-iut}\right]_{-\infty}^{+\infty} + iu\int_{-\infty}^{+\infty} f(t)e^{-iut}\,dt$	Integration by parts with $u = e^{-iut}$, $dv = f'(t)\,dt$.
$\mathcal{F}\{f'(t)\} = iu\,\mathcal{F}\{f(t)\}$	The boundary term vanishes at $t = \pm\infty$.
$\mathcal{F}\{f^{(n)}(t)\} = (iu)^n\mathcal{F}\{f(t)\}$	Induction hypothesis.
$\mathcal{F}\{f^{(n+1)}(t)\} = \mathcal{F}\{(f^{(n)}(t))'\} = iu\,\mathcal{F}\{f^{(n)}(t)\}$	Apply the $n = 1$ result to the function $f^{(n)}$.

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Mathematical Steps	Justification
$\mathcal{F}\{f^{(n+1)}(t)\} = iu\,(iu)^n\mathcal{F}\{f(t)\} = (iu)^{n+1}\mathcal{F}\{f(t)\}$	Substitute the induction hypothesis and simplify.
$\therefore\ \mathcal{F}\{f^{(n)}(t)\} = (iu)^n\mathcal{F}\{f(t)\}$	The statement holds for all $n \geq 1$ by induction.

3.3 $\frac{d^n}{du^n}\mathcal{F}\{f\} = (-i)^n\mathcal{F}\{t^n f\}$

Mathematical Steps	Justification
$\dfrac{d^n}{du^n}\mathcal{F}\{f(t)\} = \dfrac{d^n}{du^n}\displaystyle\int_{-\infty}^{+\infty} f(t)e^{-iut}\,dt$	Start from the definition of $\mathcal{F}$.
$\dfrac{d^n}{du^n}\mathcal{F}\{f(t)\} = \displaystyle\int_{-\infty}^{+\infty} f(t)\dfrac{\partial^n}{\partial u^n}e^{-iut}\,dt$	Differentiate under the integral sign.
$\dfrac{d^n}{du^n}\mathcal{F}\{f(t)\} = \displaystyle\int_{-\infty}^{+\infty} f(t)(-it)^n e^{-iut}\,dt$	$\frac{\partial^n}{\partial u^n}e^{-iut} = (-it)^n e^{-iut}$.
$\dfrac{d^n}{du^n}\mathcal{F}\{f(t)\} = (-i)^n\displaystyle\int_{-\infty}^{+\infty} t^n f(t)e^{-iut}\,dt$	Factor out $(-i)^n$.
$\dfrac{d^n}{du^n}\mathcal{F}\{f(t)\} = (-i)^n\mathcal{F}\{t^n f(t)\}$	Definition of the Fourier transform.

3.4 $\mathcal{F}\{f\}\mathcal{F}\{g\} = \mathcal{F}\{f * g\}$

Mathematical Steps	Justification
$\mathcal{F}\{f * g\}(u) = \displaystyle\int_{-\infty}^{+\infty} (f * g)(t)e^{-iut}\,dt$	Start from the definition of the Fourier transform.
$\mathcal{F}\{f * g\}(u) = \displaystyle\int_{-\infty}^{+\infty}\left[\int_{-\infty}^{+\infty} f(\tau)g(t-\tau)\,d\tau\right]e^{-iut}\,dt$	Expand the convolution.
$\mathcal{F}\{f * g\}(u) = \displaystyle\int_{-\infty}^{+\infty} f(\tau)\left[\int_{-\infty}^{+\infty} g(t-\tau)e^{-iut}\,dt\right]d\tau$	Apply Fubini's theorem and move $f(\tau)$ outside the inner integral.
$\mathcal{F}\{f * g\}(u) = \displaystyle\int_{-\infty}^{+\infty} f(\tau)\left[\int_{-\infty}^{+\infty} g(y)e^{-iu(y+\tau)}\,dy\right]d\tau$	Substitute $y = t - \tau$, so $t = y + \tau$ and $dt = dy$.
$\mathcal{F}\{f * g\}(u) = \displaystyle\int_{-\infty}^{+\infty} f(\tau)e^{-iu\tau}\left[\int_{-\infty}^{+\infty} g(y)e^{-iuy}\,dy\right]d\tau$	Split the exponential $e^{-iu(y+\tau)}$.
$\mathcal{F}\{f * g\}(u) = \mathcal{F}\{g\}(u)\displaystyle\int_{-\infty}^{+\infty} f(\tau)e^{-iu\tau}\,d\tau$	The bracketed integral is $\mathcal{F}\{g\}(u)$.
$\mathcal{F}\{f * g\}(u) = \mathcal{F}\{f\}(u)\mathcal{F}\{g\}(u)$	The remaining integral is $\mathcal{F}\{f\}(u)$.